

Experiment ‘p_3_6_parametrised_order’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_3_6_parametrised_order

Natural language statement: null

Method signature: p_3_6_parametrised_order (i: int, j: int, k: int, strict: bool) returns (s: string)

Ensures

- if strict then if i < j < k then s == "increasing" else if i > j && j > k then s == "decreasing" else s == "neither" else if i <= j <= k then if i == j == k then s == "increasing and decreasing" else s == "increasing" else if i >= j && j >= k then s == "decreasing" else s == "neither"

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_3_6_parametrised_order (i: int, j: int, k: int, strict: bool ) returns (s: string)`

The method should respect the following contract:

ensures if strict then if i < j < k then s == "increasing" else if i > j && j > k then s
→ == "decreasing" else s == "neither" else if i <= j <= k then if i == j == k then s
→ == "increasing and decreasing" else s == "increasing" else if i >= j && j >= k
→ then s == "decreasing" else s == "neither"

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn3BYeiplW4zGzORLFaehNFG4p1F

Model: gpt-5.5-2026-04-23

Created at: 1780582688

Finish reason: stop

Usage: 446 tokens in, and 293 tokens out

```
{
  if strict {
    if i < j < k {
      s := "increasing";
    } else if i > j && j > k {
      s := "decreasing";
    } else {
      s := "neither";
    }
  } else {
    if i <= j <= k {
      if i == j == k {
        s := "increasing and decreasing";
      } else {
        s := "increasing";
      }
    } else if i >= j && j >= k {
      s := "decreasing";
    }
  }
}
```

```

    } else {
      s := "neither";
    }
  }
}

```

Final Program

```

method p_3_6_parametrised_order (i: int, j: int, k: int, strict: bool ) returns (s:
  ↪ string)
  ensures if strict then if i < j < k then s == "increasing" else if i > j && j > k
    ↪ then s == "decreasing" else s == "neither" else if i <= j <= k then if i ==
    ↪ j == k then s == "increasing and decreasing" else s == "increasing" else
    ↪ if i >= j && j >= k then s == "decreasing" else s == "neither"
{
  if strict {
    if i < j < k {
      s := "increasing";
    } else if i > j && j > k {
      s := "decreasing";
    } else {
      s := "neither";
    }
  } else {
    if i <= j <= k {
      if i == j == k {
        s := "increasing and decreasing";
      } else {
        s := "increasing";
      }
    } else if i >= j && j >= k {
      s := "decreasing";
    } else {
      s := "neither";
    }
  }
}

```

Total Token Usage

Input tokens: 446

Output tokens: 293

Reasoning tokens: 118

Sum of 'total tokens': 739

Experiment Timings

Overall Experiment started at 1780582686668, ended at 1780582723347, lasting 36679ms (36.68 seconds)

Iteration #1 started at 1780582686669, ended at 1780582723347, lasting 36678ms (36.68 seconds)